Logarithms and Logarithmic Equations

Concept

A logarithm answers: what exponent produces this number?

Definition

$\log_b x = y$ means $b^y = x$; $b > 0$, $b \neq 1$, $x > 0$.

Log laws

Product → sum; quotient → difference; power → multiplier, for positive arguments.

Change of base

$\log_b x = \ln x / \ln b$. log uses base 10; ln uses base e.

Restrictions

Every original log argument must be positive; check candidate roots.

Example 1

Solve $\log(x - 1) + \log(x + 2) = 1$.

Solution

1. Domain: $x > 1$. Combine: $\log[(x - 1)(x + 2)] = 1$.
2. $(x - 1)(x + 2) = 10 \rightarrow (x + 4)(x - 3) = 0$.
3. Candidates $-4, 3$; only $x = 3$ fits the domain.

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Example 2

Solve $5 \times 2^x = 30$.

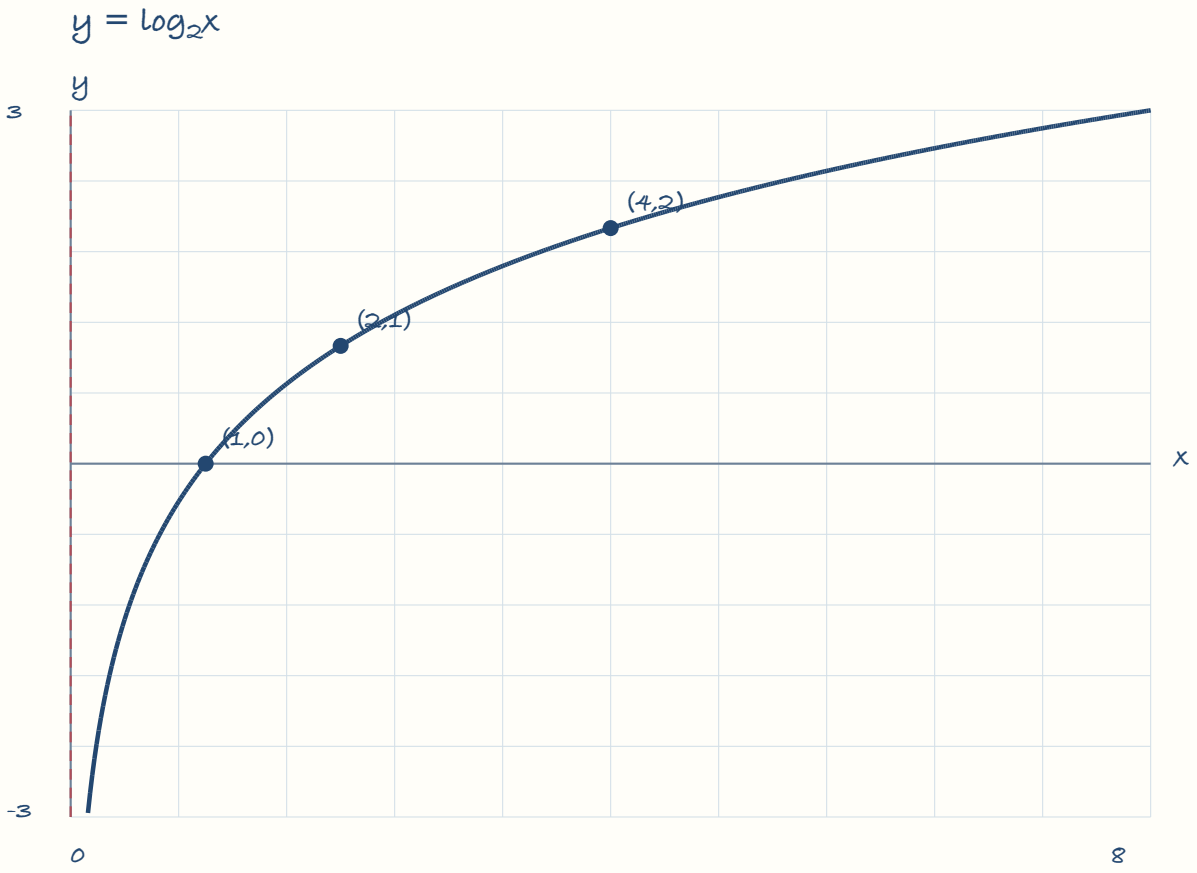
Solution

1. $2^x = 6 \rightarrow x \ln 2 = \ln 6$.

2. $x = \ln 6 / \ln 2 \approx 2.585$.

Logarithms and Logarithmic Equations

Visual notes | read the graph



The logarithm is undefined for $x \le 0$. Doubling the input adds 1 to the output.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Logarithms and Logarithmic Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Evaluate $\log_4 64$.

2. Expand $\ln(3x^2/y)$ for $x, y > 0$.

3. Solve $\ln(x - 2) = 0$.

4. Solve $3^x = 20$ exactly using logs.

5. Solve $\log_2 x + \log_2(x - 2) = 3$.

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Practice answers | check after trying

1. 3.

2. $\ln 3 + 2\ln x - \ln y$.

3. $x - 2 = e^0 = 1$, so $x = 3$.

4. $x = \ln 20 / \ln 3 \approx 2.727$.

5. $x > 2$. $x(x - 2) = 8$ gives $(x - 4)(x + 2) = 0$. Only $x = 4$ is valid.